Integer solutions of $x^4 + y^3 + z^2 + 1 = 0$: geometry, numerical data, and the absence of low-degree polynomial families over $\mathbb{Q}$

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15 October 2026

Abstract

We study the open question whether $x^4 + y^3 + z^2 + 1 = 0$ has infinitely many integer solutions. We do not settle it. We record (i) an exhaustive search, which finds 35 solutions with $x, z \geq 0$ and $|y| \leq 3.3 \cdot 10^6$, and compare the count with a Hardy–Littlewood prediction; (ii) the geometry: the surface is the Milnor fibre of the E_6 singularity, fibred over the x-line by Mordell curves into a rational elliptic surface whose Mordell–Weil lattice is E_6^*, and a Frobenius-trace computation shows that Galois fixes no nonzero vector of this lattice, so there is no section over $\mathbb{Q}(x)$ and the coordinate ring over $\mathbb{Q}$ is factorial; (iii) the nonexistence of polynomial solutions in $\mathbb{Q}[t]^3$ with $\deg x \leq 2$, and with $\deg x = 3$, $\deg y \leq 4$. These exclusions combine elementary arguments with Gröbner bases and 2-descent. Over $\overline{\mathbb{Q}}$ the surface carries many polynomial curves, including a real one-parameter family of identities of degrees $(3, 4, 6)$, so the obstruction over $\mathbb{Q}$ is arithmetic rather than geometric. We list precisely which degree patterns remain open and reduce the next one to an explicit polynomial system that we could not solve.

1 Introduction

Let S be the affine surface $x^4 + y^3 + z^2 + 1 = 0$. Are there infinitely many integer points on S? The question is stated as open in Grechuk's book [1] (see also [2]). Examples are $(0, -1, 0)$, $(1, -3, 5)$, $(6, -13, 30)$, $(11, -27, 71)$ and $(31, -147, 1501)$; a longer list is in Appendix A.

Writing $Y = -y$, the equation becomes

$$Y^3 \;=\; x^4 + z^2 + 1, \tag{1}$$

which can be read in three ways: a cube is a fourth power plus a square plus one; the Mordell equation $z^2 = Y^3 - (x^4 + 1)$ has an integral point for the parameter x; the sum of the three squares $(x^2)^2 + z^2 + 1^2$ is a cube.

Because $\frac{1}{4} + \frac{1}{3} + \frac{1}{2} > 1$, a probabilistic count predicts roughly $T^{1/4}$ solutions with $Y \leq T$ (Section 3), so infinitely many are expected. But the growth is so slow that data cannot test subtle structure, and no known method proves infinitude. The usual route is to exhibit a curve on the surface that carries infinitely many integer points: by Siegel's theorem such a curve must have genus 0 and at most two points at infinity, that is, a polynomial parametrisation or a Pell-type conic. This paper asks which of these routes are available, and why not.

*An AI model. This report was written at the request of a user who posed the problem. It has not been peer reviewed, its novelty has not been checked against the literature, and several of its statements rest on computer algebra (flagged throughout).

Results

R1. *Data* (Section 3). All solutions with $x, z \geq 0$ and $Y \leq 3\,300\,000$: there are 35. The Hardy–Littlewood prediction is 47; the deficit is discussed.

R2. *Geometry* (Section 2). S is the Milnor fibre of E_6. Over the x-line it is a rational elliptic surface with fibres $4 \times \mathrm{II} + \mathrm{IV}$ and Mordell–Weil lattice E_6^*. Traces of Frobenius at the primes 7^k show that Galois fixes no nonzero vector of E_6^* (Proposition 2.3); hence there is no section over $\mathbb{Q}(x)$ and $\mathbb{Q}[x, y, z]/(x^4 + y^3 + z^2 + 1)$ is a UFD (Corollary 2.4).

R3. *Curves over* $\overline{\mathbb{Q}}$ (Section 4). There are many polynomial curves over $\overline{\mathbb{Q}}$: an explicit one, and a real one-parameter family of degrees $(3, 4, 6)$ cut out by hyperplane sections of an embedding $S \hookrightarrow \mathbb{P}^4$.

R4. *Theorem A* (Section 5). There is no nonconstant $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ solving the equation with $\deg x \leq 2$.

R5. *Theorem B* (Section 6). There is none with $\deg x = 3$ and $\deg y \leq 4$ (forcing degrees $(3, 4, 6)$). The proof is elementary in the symmetric subcase and uses computer algebra and 2-descent on two genus-one curves otherwise.

R6. *Open cases* (Section 7). For $\deg x = 3$ and $\deg y \geq 5$ only $\deg y \in \{8, 10, 14, 16, 20, 22\}$ can occur; the first is an explicit polynomial system whose Gröbner basis we could not compute. Nothing is proved for $\deg x \geq 4$ or for Pell-type curves.

Verification status

Each result carries a tag. [E] means a complete argument that can be checked by hand, relying only on standard theorems. [C] means the proof uses a computer-algebra computation (Gröbner bases in Singular, 2-descent in PARI/GP) that we did not verify independently by hand; it is reproducible with the accompanying script (Section 9). [N] means numerical or heuristic evidence.

Statement	Tag	Where
List of 35 solutions up to $Y \leq 3.3 \cdot 10^6$	[C] (exact search)	Section 3, Appendix A
Counting heuristic, constant ≈ 0.42	[N]	Section 3
$\mathrm{MW}(\hat{E}/\mathbb{Q}(x)) = 0$, $\mathbb{Q}[S]$ is a UFD	[E]/[C] (finite field counts)	Proposition 2.3, Corollary 2.4
Real family of $(3, 4, 6)$ identities over $\overline{\mathbb{Q}}$	[N], one member [E]	Section 4
Theorem A ($\deg x \leq 2$)	[C]	Section 5
Theorem B, symmetric and $c_1 = 0$ subcases	[E]	Section 6
Theorem B, non-symmetric subcase	[C]	Section 6, Appendix B
Reduction of $\deg x = 3$, $\deg y \geq 5$ to six degrees	[E] (sketch)	Section 7

Table 1: Status of the statements.

2 Geometry of the surface

2.1 The elliptic fibration

Put $Y = -y$. For fixed x the fibre is the Mordell curve $E_x : z^2 = Y^3 - (x^4+1)$. Regard $z^2 = Y^3 + a_6(x)$ with $a_6 = -(x^4+1)$ as a Weierstrass equation over $\mathbb{Q}(x)$. Since $\deg a_6 \le 6$, the minimal elliptic surface $\pi : \hat{E} \to \mathbb{P}^1$ is rational. Its discriminant is $-432(x^4+1)^2$. At the four roots of x^4+1 the coefficient a_6 vanishes to order 1: type II. At $x=\infty$, with $t = 1/x$ and the usual scaling, $a_6 \mapsto -(t^2+t^6)$ vanishes to order 2: type IV. The Euler numbers add up: $4\cdot 2 + 4 = 12$.

The affine surface S is $\hat{E}$ minus the zero section O and the fibre at infinity $F_\infty = \Theta_0 + \Theta_1 + \Theta_2$ (the cusp of each type-II fibre is a smooth point of the total space, because $\partial_x(x^4+1) \neq 0$ there). The Shioda–Tate formula gives $\operatorname{rank} \mathrm{MW}(\hat{E}/\overline{\mathbb{Q}}(x)) = 8 - (3-1) = 6$, and by Oguiso–Shioda [3] the Mordell–Weil lattice is E_6^*. The trivial lattice $T = \langle O, F, \Theta_1, \Theta_2\rangle \subset \mathrm{Pic}(\hat{E}) \cong I_{1,9}$ is primitive of discriminant 3 with $T^\perp \cong E_6(-1)$, so

$$\mathrm{Pic}(\bar{S}) \;=\; \mathrm{Pic}(\hat{E}_{\overline{\mathbb{Q}}})/\langle O, \Theta_0, \Theta_1, \Theta_2\rangle \;\cong\; E_6^*, \qquad \overline{\mathbb{Q}}[S]^* = \overline{\mathbb{Q}}^*.$$

There are 126 sections whose coordinates (Y, z) are polynomials in x of degrees at most $(2,3)$: the 72 roots and the 54 minimal vectors of E_6^*. They are polynomial curves over $\overline{\mathbb{Q}}$; for instance $Y = \omega^j$, $z = \pm i x^2$ ($\omega^3 = 1$) are six of the 54.

Remark 2.1. The weighted projective compactification $X_{12} \subset \mathbb{P}(1,3,4,6)$ of S has $-K_X = \mathcal{O}(2)$ and boundary $\{w = 0\} \sim \mathcal{O}(1)$, so $K_X + D$ is anti-ample. This is the usual picture of a Milnor fibre of an ADE singularity as a rational affine surface of log Kodaira dimension $-\infty$, where one expects Zariski-dense integral points in the absence of obstructions. We do not use this.

2.2 Galois action on the lattice

For a prime power $q = p^k$, $p \ge 5$, put

$$T_q \;=\; \frac{1}{q} \sum_{x,Y \in \mathbb{F}_q} \chi\big(Y^3 - x^4 - 1\big), \tag{2}$$

where χ is the quadratic character.

Lemma 2.2. *$\#S(\mathbb{F}_q) = q^2 + q\,T_q$, and $T_q = \mathrm{Tr}\big(\mathrm{Frob}_q \mid \mathrm{Pic}(\bar{S}) \otimes \mathbb{Q}_\ell\big)$.*

Proof. [E] Since $\hat{E}$ is rational, $\#\hat{E}(\mathbb{F}_q) = 1 + q^2 + q\,\mathrm{Tr}(\mathrm{Frob}_q \mid \mathrm{NS})$, and $\mathrm{NS} \otimes \mathbb{Q}_\ell = T \oplus V$ with $V \cong \mathrm{Pic}(\bar{S}) \otimes \mathbb{Q}_\ell$. The Frobenius trace on T is $2 + \tau$, where $\tau \in \{0, 2\}$ is the trace on $\langle \Theta_1, \Theta_2\rangle$ ($\tau = 2$ exactly when these components are defined over $\mathbb{F}_q$, that is, when $q \equiv 1 \bmod 4$). The boundary $O \cup F_\infty$ has $(q+1) + (q+1+\tau q) - 1$ points. Subtracting, $\#S(\mathbb{F}_q) = q^2 + q\,\mathrm{Tr}(\mathrm{Frob}_q \mid V)$. Finally $\#S(\mathbb{F}_q) = \sum_{x,Y}(1 + \chi(Y^3 - x^4 - 1)) = q^2 + qT_q$. □

Proposition 2.3. *[C] The traces T_p for the 194 primes $5 \le p \le 1200$ take the values $-2, 0, 2, 6$ with frequencies $61, 100, 23, 10$ and mean -0.08. At $p = 7$ one has $T_{7^k} = -2, -2, -2, 6$ for $k = 1, 2, 3, 4$. Consequently Frob_7 has order 4 on the six-dimensional space V and no eigenvalue 1, and*

$$\mathrm{MW}(\hat{E}/\mathbb{Q}(x)) \;=\; 0.$$

Proof. The Galois action on the finitely generated group $\mathrm{NS}(\bar{\hat{E}})$ factors through a finite group, so Frobenius eigenvalues are roots of unity. Since $T_{7^4} = 6 = \dim V$, $\mathrm{Frob}_7^4 = 1$. The dimension

of the Frob_7-fixed space is $\frac{1}{4}\sum_{j=0}^{3}\mathrm{Tr}(\mathrm{Frob}_7^j) = \frac{1}{4}(6-2-2-2) = 0$. The prime 7 is of good reduction (the bad primes are $2, 3$). The specialisation map $\mathrm{NS}(\hat{E}_{\overline{\mathbb{Q}}}) \to \mathrm{NS}(\hat{E}_{\mathbb{F}_7})$ is an injective isometry between unimodular lattices of rank 10, hence an isomorphism compatible with the decomposition group. A section over $\mathbb{Q}(x)$ would give a nonzero Frob_7-invariant vector of V. □

The finite-field counts were computed by the script of Section 9; the argument above is then a finite verification. The Chebotarev average over all primes (-0.08 for $p \le 1200$) is consistent with a zero fixed space, as it must be.

Corollary 2.4. *[C] (i) The elliptic surface has no section over $\mathbb{Q}(x)$ other than O; in particular no family of solutions is parametrised by x. (ii) $\mathrm{Pic}(S_{\mathbb{Q}}) \hookrightarrow \mathrm{Pic}(\bar{S})^{\mathrm{Gal}} = 0$, hence $\mathbb{Q}[x,y,z]/(x^4+y^3+z^2+1)$ is a UFD. Every irreducible curve on $S_{\mathbb{Q}}$ is the zero set of a single polynomial.*

Proof. (i) is Proposition 2.3. (ii) The Hochschild–Serre sequence gives $\mathrm{Pic}(S_{\mathbb{Q}}) \hookrightarrow \mathrm{Pic}(\bar{S})^{\mathrm{Gal}}$ because $H^1(\mathrm{Gal}, \overline{\mathbb{Q}}^*) = 0$ and $\overline{\mathbb{Q}}[S]^* = \overline{\mathbb{Q}}^*$. A smooth affine variety with trivial class group has factorial coordinate ring. □

2.3 What a family has to look like

Proposition 2.5 (Siegel)**.** *An irreducible affine curve over $\mathbb{Q}$ with infinitely many integral points has geometric genus 0 and at most two places at infinity.*

Hence an infinite family of integer solutions arising from a curve in S is a polynomial parametrisation (one place at infinity; it is then rational over $\mathbb{Q}$, and the curve is $\mathbb{A}^1_{\mathbb{Q}}$), or a Pell-type conic (two places, conjugate over a real quadratic field or rational). A nonconstant polynomial solution $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ gives infinitely many integer solutions, since the polynomials are integer-valued on an arithmetic progression. If x is constant the polynomials y, z satisfy $y^3 + z^2 = \text{const} \neq 0$ and are constant; so below x is nonconstant.

Remark 2.6. The fibration by Y has fibres $x^4 + z^2 = Y^3 - 1$, genus-one quartics with real points only for $Y \ge 1$; it has no section even over $\mathbb{R}(Y)$. The y-fibration therefore gives no families.

3 Data and heuristic

3.1 Search

For each Y, put $c = Y^3 - 1$ and loop over $0 \le x \le c^{1/4}$, testing whether $c - x^4$ is a perfect square. Two cheap filters (squares modulo 64 and modulo 63, using precomputed tables of x^4 mod 64 and mod63) discard most pairs before an exact 128-bit integer square root. The number of pairs is $\sum_{Y \le T} Y^{3/4} \approx \frac{4}{7}T^{7/4}$; for $T = 3.3 \cdot 10^6$ this is about $1.5 \cdot 10^{11}$ before filtering, a few minutes on one core. (A sum-of-two-squares method that factors $Y^3 - 1 = (Y-1)(Y^2+Y+1)$ would be quasi-linear in T; we did not implement it.)

All hits were rechecked in big-integer arithmetic. There are 35 solutions with $x, z \ge 0$ and $1 \le Y \le 3.3 \cdot 10^6$, all listed in Appendix A. The first 17 ($Y \le 90\,891$) agree with the list supplied with the problem; the others are new to us. The value $y = -167051$ carries two solutions, $x = 479$ and $x = 7365$.

3.2 Heuristic

The expected number of $(x,z) \in \mathbb{Z}^2_{\geq 0}$ with $x^4+z^2+1 = Y^3$ is $\sigma_\infty(Y)\prod_p \sigma_p$ with $\sigma_\infty(Y) = \frac{K}{2}Y^{-3/4}$, where $K = \int_0^1 (1-u^4)^{-1/2}du = \Gamma(\tfrac14)^2/(4\sqrt{2\pi}) \approx 1.3110$. Summing over $Y \leq T$ gives

$$N(T) \;\approx\; 2K\,T^{1/4}\prod_p \sigma_p, \tag{3}$$

where $\sigma_p = \lim p^{-2k}\#\{(x,z,Y) \bmod p^k : \; x^4+z^2+1 \equiv Y^3\}$. One finds $\sigma_2 = \frac12$ (computed modulo 2^k for $k = 4,\dots,9$) and $\sigma_3 = \frac{11}{9}$ (modulo 3^k, $k = 2,\dots,5$). For $p \geq 5$ the surface is smooth modulo p, so $\sigma_p = \#S(\mathbb{F}_p)/p^2 = 1 + T_p/p$. The Euler product is only conditionally convergent: it differs by an absolutely convergent factor from the value at $s = 1$ of the L-function of the Artin representation V, which has no trivial constituent. Truncated at $p \leq 3000$ it equals 0.4245.

T	observed	predicted (3)
90 891	17	19.3
1 000 000	27	35.2
2 642 245	34	44.9
3 300 000	35	47.4

Table 2: Solutions with $x,z \geq 0$ and $Y \leq T$.

[N] The observed counts lie about 25% below the prediction for $T \geq 10^6$; the Poisson standard deviation at 47 is about 7, so the deficit is roughly 1.7 standard deviations, which we regard as consistent given the conditional convergence of the product and unknown secondary terms. The statement that matters here is qualitative: $N(T) \asymp T^{1/4}$ is far too sparse for density arguments to prove infinitude.

4 Polynomial curves over $\overline{\mathbb{Q}}$

4.1 An explicit curve

With $a = 2^{2/3}$,

$$(x,y,z) \;=\; \bigl(t^3,\; -t^4 - 2^{2/3},\; \sqrt{3}\,(2^{1/3}t^4+1)\bigr) \tag{4}$$

satisfies $x^4+y^3+z^2+1 = 0$ identically. [E] Indeed $(t^4+a)^3 = t^{12}+3at^8+3a^2t^4+4$ and $z^2 = 3at^8+3a^2t^4+3$ (using $a^3 = 4$), so $Y^3 - z^2 - x^4 = 1$. This curve is defined over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ and has no rational points.

4.2 Hyperplane sections

The functions $1, x, x^2, Y, z$ embed S in $\mathbb{P}^4$ as the complete intersection

$$u_1^2 = u_0u_2, \qquad u_3^3 = u_0\,(u_2^2+u_4^2+u_0^2)$$

of a quadric cone and a cubic, via $(1 : x : x^2 : Y : z)$. A hyperplane section not through the point at infinity is a curve

$$z \;=\; c_1Y + c_2x^2 + c_3x + c_4. \tag{5}$$

Its plane model $G(x,Y) = Y^3 - (c_1Y + c_2x^2 + c_3x + c_4)^2 - x^4 - 1 = 0$ has Newton polygon the triangle with vertices $(0,0),(4,0),(0,3)$, which has three interior lattice points: generic

members have genus 3. A member of geometric genus 0 has three nodes, a codimension-3 condition in the four-dimensional family, so the rational members form a curve $\mathfrak{B} \subset \mathbb{A}^4(c_1,\dots,c_4)$. Such a member has exactly one place at infinity, where (x, Y, z) have pole orders $(3,4,6)$, so it is parametrised by polynomials of degrees $(3,4,6)$ (details in Appendix C).

[N] Numerical solution of the defining system gives real points of $\mathfrak{B}$. For $c_2 \approx 1.5411$ one finds $(c_1, c_3, c_4) \approx (-2.36822, -1.62357, 2.54626)$ and the real polynomial identity

$$\begin{aligned} x(t) &= t^3 + 2.47219\,t - 0.52273,\\ Y(t) &= 1.49999\,t^4 + 3.32230\,t^2 - 1.78682\,t + 1.23334,\\ z(t) &= 1.5411\,t^6 + 4.06747\,t^4 - 3.23473\,t^3 + 1.55084\,t^2 - 3.76529\,t + 0.89522, \end{aligned}$$

for which the coefficients of $Y^3 - z^2 - x^4 - 1$ are below $6 \cdot 10^{-14}$. The curve (4) is the member $(c_1, c_2, c_3, c_4) = (\sqrt{3}\,2^{1/3}, 0, 0, -\sqrt{3})$. So over $\overline{\mathbb{Q}}$, and over $\mathbb{R}$, polynomial solutions of degrees $(3,4,6)$ form a one-parameter family. The question is whether $\mathfrak{B}$ has a rational point.

5 Polynomial solutions with $\deg x \le 2$

Lemma 5.1 (trace lemma). *Let $(x(t), Y(t), z(t)) \in \mathbb{Q}[t]^3$ be a solution of (1) with $d = \deg x \ge 1$, injective on a dense open set. Let $t_1, \dots, t_d$ be the roots of $x(t) = x$ over $\overline{\mathbb{Q}(x)}$ and $P_i = (Y(t_i), z(t_i)) \in E_x$. Then $P_1 + \cdots + P_d = O$ in the group of E.*

Proof. The zero-cycle $\sum [P_i]$ on the generic fibre $E_{\mathbb{Q}(x)}$ is Galois-stable, so its sum lies in $E(\mathbb{Q}(x))$, which is $\{O\}$ by Corollary 2.4. □

Theorem 5.2 (A). *[C] There is no nonconstant $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ solving $x^4 + y^3 + z^2 + 1 = 0$ with $\deg x \le 2$.*

Proof. If $\deg x = 1$ the solution is a section over $\mathbb{Q}(x)$, impossible. Let $\deg x = 2$. By Lemma 5.1, $P_2 = -P_1$, so $Y(t_1) = Y(t_2)$ and $z(t_1) = -z(t_2)$. Hence $Y = Y_0(x(t))$ with $Y_0 \in \mathbb{Q}[x]$, and $z = (2at + b)\, g(x(t))$ with $g \in \mathbb{Q}[x]$, where $x = at^2 + bt + c$ and $(2at + b)^2 = 4a(x - x_*)$ for the critical value $x_* \in \mathbb{Q}$. The equation becomes the polynomial identity

$$Y_0(x)^3 - x^4 - 1 \;=\; \lambda\,(x - x_*)\, g(x)^2, \qquad \lambda = 4a \in \mathbb{Q}^*. \tag{6}$$

Let $m = \deg Y_0$. The right side has odd degree, so m is odd and $m \ge 3$ (for $m \le 2$ the left side has degree 4 or 6), and $\deg g = (3m - 1)/2$. The three terms Y_0^3, $-(x^4 + 1)$, $-\lambda(x - x_*)g^2$ sum to zero and are pairwise coprime (a common root would be a root of $x^4 + 1$ equal to $x_* \in \mathbb{Q}$, which is impossible). The Mason–Stothers inequality gives $3m \le N_0 - 1 \le m + 4 + 1 + \frac{3m-1}{2} - 1$, i.e. $m \le 7$. So $m \in \{3, 5, 7\}$.

Since g divides both $F = Y_0^3 - x^4 - 1$ and F', it divides $R_0 := Y_0 F' - 3Y_0' F = 3Y_0'(x^4 + 1) - 4x^3 Y_0$, of degree $m + 3$ and leading coefficient $(3m - 4)y_m$.

- $m = 7$: $\deg g = 10 = \deg R_0$, so, normalising g to be monic, $g = R_0/(17y_7)$. Then (6) is a system of 21 equations in the 9 unknowns $y_0, \dots, y_7, x_*$ (with $y_7 \neq 0$). Its Gröbner basis over $\mathbb{Q}$ is $\{1\}$.
- $m = 5$: $\deg R_0 = 8$, so $R_0 = 11y_5(x - r)g$ for some r. This gives 16 equations in the 8 unknowns $y_0, \dots, y_5, r, x_*$. Gröbner basis $\{1\}$.

- $m = 3$: write $F = y_3^3(x - x_*)G^2$ with G monic of degree 4. This is 9 equations in the 9 unknowns $y_0, \dots, y_3, g_0, \dots, g_3, x_*$; the solution set over $\overline{\mathbb{Q}}$ is finite of degree 108 (with multiplicity). Elimination gives the minimal polynomial of x_*: a polynomial of degree 36 in x_*^4 (degree 9 in x_*^4) which is irreducible over $\mathbb{Q}$. So $x_* \notin \mathbb{Q}$.

In every case there is no rational solution. □

Remark 5.3. For $m = 3$ complex solutions do exist: a numerical search finds, for instance, $x_* \approx -2.1956$ and $x_* \approx \pm 2.1956\, i$ (the system is invariant under $x \mapsto ix$, so $x_* \approx 2.1956\, i^k$). So over $\overline{\mathbb{Q}}$ there are polynomial curves with $\deg x = 2$, but none is defined over $\mathbb{Q}$.

6 Polynomial solutions with $\deg x = 3$

Lemma 6.1. *Let $(x(t), Y(t), z(t)) \in \mathbb{Q}[t]^3$ be a solution with $\deg x = 3$, injective on a dense open set. Then there are $m, n \in \mathbb{Q}[x]$ with $z(t) = m(x(t))Y(t) + n(x(t))$. The plane curve $G(x, Y) = Y^3 - (mY + n)^2 - x^4 - 1$ is the characteristic polynomial of Y over $\mathbb{Q}(x)$, it is irreducible, and the curve is its normalisation.*

Proof. [E] By Lemma 5.1 the three points P_i lie on a line $z = mY + n$ with $m, n \in \mathbb{Q}(x)$ (the line is Galois-stable). The $Y(t_i)$ are the roots of $G(x, \cdot)$, so with e_j the elementary symmetric functions of the $Y(t_i)$ we get $e_1 = m^2$, $e_2 = -2mn$, $e_3 = n^2 + x^4 + 1$. As $\mathbb{Q}[t]$ is free of rank 3 over $\mathbb{Q}[x]$ with basis $1, t, t^2$, each $e_j \in \mathbb{Q}[x]$. Then $m^2 \in \mathbb{Q}[x]$ gives $m \in \mathbb{Q}[x]$, and $n^2 = e_3 - x^4 - 1 \in \mathbb{Q}[x]$ gives $n \in \mathbb{Q}[x]$. The degree $[\mathbb{Q}(t) : \mathbb{Q}(x)] = 3$ is prime, so $\mathbb{Q}(x, Y)$ is $\mathbb{Q}(x)$ or $\mathbb{Q}(t)$; in the first case the curve factors through a section, which does not exist. □

Lemma 6.2. *In Lemma 6.1, $m \neq 0$.*

Proof. [E] If $m = 0$ then $Y(t)^3 = f(x(t))$ with $f = n^2 + x^4 + 1$, which has degree 4 and satisfies $f \geq 1$ on $\mathbb{R}$. As $f \circ x$ is a cube, each root ρ of f satisfies: either its multiplicity in f is divisible by 3, or $x(t) - \rho$ is a cube of a linear polynomial. Since $\deg f = 4 \not\equiv 0 \bmod 3$, some root ρ falls under the second case. Then $x(t) = x_3(t - a)^3 + \rho$ with $a = -x_2/(3x_3) \in \mathbb{Q}$, so $\rho = x(a) \in \mathbb{Q}$ and $f(\rho) = 0$, contradicting $f \geq 1$. □

Theorem 6.3 (B)**.** *There is no nonconstant solution in $\mathbb{Q}[t]^3$ with $\deg x = 3$ and $\deg y \leq 4$. Such a solution would necessarily have degrees $(3, 4, 6)$.*

Proof of Theorem 6.3. Write $Y = -y$. If $\deg Y \leq 3$ then $\deg Y^3 \leq 9 < 12$, so the leading terms of z^2 and x^4 would cancel, impossible since both are positive over $\mathbb{R}$; hence $\deg Y = 4$ and $\deg z = 6$. By Lemmas 6.1 and 6.2, since $\deg e_1 \leq 4/3$ and $\deg e_2 \leq 8/3$, $m = c_1 \in \mathbb{Q}^*$ is constant and $n = c_2x^2 + c_3x + c_4$ is quadratic. So the curve is a rational member of the family (5).

Symmetric case, $c_3 = 0$. [E] Then G is even in x, so $\sigma : (x, Y) \mapsto (-x, Y)$ preserves the curve and lifts to an involution $\tilde{\sigma}$ of its normalisation $\mathbb{P}^1_t$, with $x \circ \tilde{\sigma} = -x$ and $Y \circ \tilde{\sigma} = Y$. It fixes the unique point at infinity, so after a translation $\tilde{\sigma}(t) = -t$, whence x is odd and Y is even. The roots of $x(t) = 0$ are 0 and $\pm t_+$, and $Y(t_+) = Y(-t_+)$. Thus $G(0, Y) = Y^3 - (c_1Y + c_4)^2 - 1$ has a multiple root $Y_0 \in \mathbb{Q}$ (a triple root if $t_+ = 0$). Put $z_0 = c_1Y_0 + c_4$. Then $z_0^2 = Y_0^3 - 1$ and $3Y_0^2 = 2c_1z_0$. So $(Y_0, z_0) \in E_0(\mathbb{Q})$ with $E_0 : z^2 = Y^3 - 1$, a curve of rank 0 whose only rational points are O and $(1, 0)$. If $z_0 = 0$ then $3Y_0^2 = 0$, contradicting $Y_0 = 1$.

The case $c_1 = 0$ is excluded by Lemma 6.2.

Non-symmetric case, $c_3 \neq 0$. [C] The remaining argument is a computation, summarised in Table 3; the system is derived in Appendix B. Over $\overline{\mathbb{Q}}$ normalise $x = t^3 + st + x_0$ and write the conditions $e_1 = c_1^2$, $e_2 = -2c_1 n$, $e_3 = n^2 + x^4 + 1$ in the unknowns $(c_1, \dots, c_4, s, w_1, w_4)$. Elimination of (s, w_1, w_4) in Singular modulo two 31-bit primes, followed by rational reconstruction, yields seventeen generators of degrees 6–9 which we take to generate the ideal of $\mathfrak{B}$ over $\mathbb{Q}$: the lifts coincide at both primes and vanish at independent numerical solutions, but this is not a pure characteristic-0 computation. The curve $\mathfrak{B}$ has degree $108 = 24 + 84$, where the 24 comes from $c_3 = 0$ and the 84 from $\mathfrak{B}^{\mathrm{ns}}$ (the part with $c_3 \neq 0$); the saturation of the ideal by the product of the two component ideals is the unit ideal, so there is no further component.

The group $\Gamma = \mu_4 \times \mu_3 \times \mu_2$ acting by $(x, Y, z) \mapsto (\zeta_4 x, \zeta_3 Y, \pm z)$ acts on $\mathfrak{B}^{\mathrm{ns}}$, and $\mathfrak{B}^{\mathrm{ns}}/\Gamma$ is the plane septic $F(R, Q) = 0$ in $R = c_2^2$, $Q = c_4^2$, of geometric genus 0. One checks that the map $\mathfrak{B}^{\mathrm{ns}} \to \{F = 0\}$ has degree $24 = |\Gamma|$. The septic has a parametrisation over $\mathbb{Q}$ of degree 7,

$$R(s) = \frac{\varkappa_R \, L_R(s)\, C_R(s)^2}{L_Z(s) M_Z(s) N_Z(s)^2}, \qquad Q(s) = \frac{\varkappa_Q \, L_Q(s)\, C_Q(s)^2}{L_Z(s) M_Z(s) N_Z(s)^2},$$

with L_R, L_Q, L_Z linear, C_R, C_Q cubic and M_Z, N_Z quadratic (primitive integer polynomials as returned by PARI), and $\varkappa_R = 1/1627800$, $\varkappa_Q = -11/1627800$. The singular locus of F consists of two $\mathbb{Q}$-irreducible components of degrees 8 and 5, so F has no rational singular point; thus every rational point of $F = 0$ comes from $s \in \mathbb{Q}$.

A rational point of $\mathfrak{B}^{\mathrm{ns}}$ has $c_2, c_4 \in \mathbb{Q}$, so $R(s)$ and $Q(s)$ are squares in $\mathbb{Q}$ for some $s \in \mathbb{Q}$. Modulo squares, $R(s) \equiv \varkappa_R L_R L_Z M_Z$ and $Q(s) \equiv \varkappa_Q L_Q L_Z M_Z$, so s is a rational point of both genus-one curves

$$C_R : \; v^2 = \varkappa_R \, L_R L_Z M_Z, \qquad C_Q : \; v^2 = \varkappa_Q \, L_Q L_Z M_Z.$$

Their Jacobians are $y^2 = x^3 - 24x - 32$ (conductor 1152, $j = 3456$) and $y^2 = x^3 - x^2 - 13x - 11$ (conductor 384, $j = 16000/3$); both have rank 0 and torsion $\mathbb{Z}/2$ (PARI `ellrank`, lower and upper bounds 0). Each of C_R, C_Q has the two rational branch points as rational points, so $C_R(\mathbb{Q})$ and $C_Q(\mathbb{Q})$ are exactly the points over s_{L_R}, s_{L_Z} and s_{L_Q}, s_{L_Z}. A common s must be s_{L_Z}, a pole of R and Q ($c_2, c_4 = \infty$, not allowed). As a direct check, $(R, Q)(s_{L_R}) = (0, 3)$ and $(R, Q)(s_{L_Q}) = (-1, 0)$, and 3 and -1 are not rational squares. The slices $c_2 = 0$ and $c_4 = 0$ of $\mathfrak{B}^{\mathrm{ns}}$ decompose over $\mathbb{Q}$ into components of degrees $72 + 6$ and $72 + 2$, none of degree 1. Hence $\mathfrak{B}^{\mathrm{ns}}(\mathbb{Q}) = \varnothing$. □

Remark 6.4 (cross-checks). The lifted generators vanish at the numerical non-symmetric solution of Section 4 (to $3 \cdot 10^{-9}$, against values of size 10^2 at a random point) and at the exact member $(\sqrt{3}\, 2^{1/3}, 0, 0, -\sqrt{3})$ (to $2 \cdot 10^{-13}$). They also vanish, to about 10^{-46}, on the symmetric component parametrised by $c_4 = \frac{r^2+3}{2r}$, $c_1^3 = -\frac{(r^2+9)^2}{8r}$, with c_2^2 solving a quadratic; this component, analysed by hand above, is thus the $c_3 = 0$ part of the ideal.

7 What remains open

7.1 $\deg x = 3$ with $\deg y \geq 5$

Proposition 7.1. *[E] (sketch) Let $(x, Y, z) \in \mathbb{Q}[t]^3$ be a solution with $\deg x = 3$ and $d = \deg Y \geq 5$. Then $d \in \{8, 10, 14, 16, 20, 22\}$.*

Sketch. (a) *Even.* If d were odd then $\deg(Y^3 - z^2) = 3d > 12$, but $Y^3 - z^2 = x^4 + 1$ has degree 12. (b) *At most* 22. By the Davenport–Stothers inequality [6, 7, 9], $\deg(Y^3 - z^2) \geq d/2 + 1$, so

Object	Computed
Severi curve $\mathfrak{B} \subset \mathbb{A}^4$ (elimination mod 2^{31}-size primes, lifted)	dimension 1, degree 108; 17 generators of degrees 6–9
Components	$c_3 = 0$: degree 24; $c_3 \neq 0$: degree 84; no others
$\mathfrak{B}^{\mathrm{ns}}/\Gamma$	plane septic in (c_2^2, c_4^2), genus 0, $\mathbb{Q}$-parametrisation of degree 7
Singular points of the septic	$\mathbb{Q}$-irreducible components of degrees 8 and 5
C_R, Jacobian	$y^2 = x^3 - 24x - 32$, conductor 1152, rank 0, torsion $\mathbb{Z}/2$
C_Q, Jacobian	$y^2 = x^3 - x^2 - 13x - 11$, conductor 384, rank 0, torsion $\mathbb{Z}/2$
Values at rational branch points	$(R, Q) = (0, 3)$ at s_{L_R}, $(-1, 0)$ at s_{L_Q}
Slices $c_2 = 0$, $c_4 = 0$ of $\mathfrak{B}^{\mathrm{ns}}$	degrees $72 + 6$ and $72 + 2$

Table 3: Computations in the non-symmetric case of Theorem 6.3.

$12 \geq d/2 + 1$, i.e. $d \leq 22$. (Here Y and z have no common root, because a common root would force a critical value of x to be a root of $x^4 + 1$, and the critical values are roots of a rational quadratic.) (c) *Not divisible by* 3. The curve has a single place at infinity, totally ramified over $x = \infty$. If $3 \mid d$, the leading term of Y is $\eta x^{d/3}$ with $d/3$ an integer, and the Newton polygon edge polynomial of G (in η, one of $\eta^3 - n_\nu^2$, $\eta^2(\eta - m_\mu^2)$, $\eta^3 - 2m_\mu n_\nu \eta$ or $\eta^3 - (m_\mu \eta + n_\nu)^2$) would have to have a triple root. None does. Thus $d \in \{8, 10, 14, 16, 20, 22\}$. □

Write $d = 2\nu$ with $\nu \in \{4, 5, 7, 8, 10, 11\}$. Then $\deg n = \nu$ and $\deg m \leq \nu/3$, and in the notation of Appendix B the unknowns are s, x_0 and the coefficients of β, γ, m. For $\nu = 4$ ($d = 8$, $\deg z = 12$, $\deg m \leq 1$) the identity $4m^2 e_3 - e_2^2 - 4m^2(x^4 + 1) = 0$ is a system of 11 equations in 10 unknowns of degree up to 8 (`deg3 2 2 1` in the script). Dimension counts suggest it has no solution: the class of such curves has expected dimension $3 - \nu < 0$ (Appendix C). We did not prove this. A Gröbner basis modulo 32003 (`std` and `slimgb`, two runs) did not terminate in about 14 minutes on one core, and a floating-point Newton search returned only badly scaled large-coefficient near-solutions which did not survive refinement; we discard them.

7.2 Other open cases

- $\deg x \geq 4$. The relevant classes (for $\deg x = 3k$ the class kF_z) have families of dimension about k, so more room, not less; nothing is proved.
- Pell-type curves (two places at infinity). For example the fibrations $(z \pm ix^2)/(Y - \omega^j)$ have fibres isomorphic to $\mathbb{G}_m$ but are defined only over $\mathbb{Q}(i, \omega)$ and are permuted by Galois, so they do not descend to $\mathbb{Q}$.
- Any mechanism not coming from a curve (for instance an infinite group of automorphisms or a direct arithmetic argument), none of which is known for this surface.

8 Discussion

Over $\overline{\mathbb{Q}}$ the surface is covered by rational curves: the 126 sections, the explicit curve (4), the one-parameter family of Section 4. Over $\mathbb{Q}$ every low-degree candidate we could settle is excluded for an arithmetic reason. The Galois module $E_6^* = \mathrm{Pic}(\bar{S})$ has no fixed vector (Proposition 2.3), so pencils of $\mathbb{A}^1$'s on S are permuted by Galois and none is defined over $\mathbb{Q}$; the curves in the hyperplane family have trivial class in $\mathrm{Pic}(S)$, and Theorem 6.3 shows that the Severi curve parametrising them has no rational points, the obstruction being the Mordell curve $y^2 = x^3 - 1$ in the symmetric branch and two rank-zero genus-one curves in the other.

This does not make infinitude unlikely; the counting heuristic of Section 3 says the opposite. It says that if infinitely many solutions exist, they do not come from a polynomial identity of the simplest kind. What a proof would need is either a rational point on one of the remaining moduli spaces of rational curves (higher degree in x, or Pell-type), or an idea that does not go through curves, as for $x^3 + y^3 + z^3 = 3$.

9 Reproducibility

All computations are in one script, `x4y3z2_toolkit.py`, written for Python 3 with `numpy` and `sympy`, a C++ compiler for the search, Singular 4.3.2 and PARI/GP for the algebra (tested on Ubuntu 24.04). Stages:

Command	Content
`search` $Y_0\,Y_1$	exact search (OpenMP, 128-bit), rechecked in big integers
`traces`	T_p, T_{7^k}, rank argument, local densities, (3)
`sym`	the explicit curve, the symmetric-case algebra, $y^2 = x^3 - 1$ check
`elim`	Severi-curve elimination modulo two primes (about 4 minutes each)
`nonsym`	lifting, component degrees, septic parametrisation, C_R, C_Q, slices
`xdeg2`	Theorem A
`deg3` $B\,C\,\mu$	writes and runs the $\deg x = 3$ system for a given degree pattern

Bugs found and fixed while preparing the script: a Singular ideal comparison (==) that is a type error, a multi-line PARI `foreach` that does not parse in a script, and a stage that silently produced no output when its subprocess was killed (the stage now checks for the output file and is resumable).

A The solutions

All solutions with $x, z \geq 0$ and $Y = -y \leq 3.3 \cdot 10^6$; the sign variants $(\pm x, y, \pm z)$ are also solutions. Each was verified in exact arithmetic.

#	x	y	z	#	x	y	z
1	0	−1	0	19	7365	−167051	41465695
2	1	−3	5	20	7081	−273651	134082973
3	6	−13	30	21	1560	−337501	196055550
4	11	−27	71	22	456	−392881	246258588
5	31	−147	1501	23	3895	−421715	273439443
6	51	−203	1265	24	17009	−566691	313509803
7	29	−5507	408669	25	21279	−609971	148069427

#	x	y	z	#	x	y	z
8	1386	−15661	388458	26	7692	−698113	580286880
9	288	−16957	2206566	27	11466	−914701	864884958
10	1669	−21531	1490663	28	36845	−1229283	121085581
11	2239	−29931	1297207	29	16194	−1435213	1699273710
12	855	−38051	7386395	30	11952	−1564573	1951792590
13	2761	−47171	6844587	31	48047	−1786587	611029361
14	2237	−57147	12711719	32	2531	−1987571	2802094917
15	1103	−84027	24326849	33	55439	−2523387	2573193569
16	2859	−85547	23648369	34	24744	−2579377	4097097156
17	949	−90891	27387137	35	62671	−2716443	2149031185
18	479	−167051	68276487				

B The system for $\deg x = 3$

Over $\overline{\mathbb{Q}}$ normalise $x = t^3 + st + x_0$ (affine change of t; the constants c_i below are intrinsic). Let $u = x - x_0$; the roots t_i of $x(t) = x$ have power sums $P_0 = 3$, $P_1 = 0$, $P_2 = -2s$, $P_k = -sP_{k-2} + uP_{k-3}$. For a polynomial $Y = \alpha + \beta t + \gamma t^2$ with $\alpha, \beta, \gamma \in \overline{\mathbb{Q}}[x]$ one has $\operatorname{Tr} Y = 3\alpha - 2s\gamma$, which must equal m^2.

For the $(3, 4, 6)$ case $m = c_1$ is constant, which forces the t^3 coefficient of Y to vanish. Put $W = Y - c_1^2/3 = \alpha + \beta t + \gamma t^2$ with $\operatorname{Tr} W = 0$, so $\alpha = 2s\gamma/3$, and $\beta = w_4 u + w_1$ where w_4 is the leading coefficient of Y and w_1 the coefficient of t. The characteristic polynomial of W is $T^3 + pT - N$ with

$$p = s\beta^2 - 3u\beta\gamma - \tfrac{1}{3}s^2\gamma^2, \qquad N = N(W) = \tfrac{1}{3}\operatorname{Tr}(W^3).$$

Comparing with $G(x, Y)$ (shift $Y = W + c_1^2/3$) gives

$$p = -2c_1\big(n + \tfrac{c_1^3}{6}\big), \qquad N = x^4 + \big(n + \tfrac{c_1^3}{3}\big)^2 + 1 - \tfrac{c_1^6}{27}, \qquad n = c_2x^2 + c_3x + c_4.$$

The first relation determines n from p (so c_3, c_4 in terms of s, x_0, w_1, γ, c_1 and c_2), and the second is a polynomial identity in u of degree 4. Its u^4 coefficient reads $(3\gamma w_4 - sw_4^2)^2 = 4c_1^2(w_4^3 - 1)$; with $c_2 := \pm\sqrt{w_4^3 - 1}$ (the leading coefficient of n) this is $3\gamma w_4 = sw_4^2 + 2c_1c_2$. The u^3 coefficient is linear in x_0 and gives $4c_1w_4x_0 = w_1\big(c_1(3 + c_2^2) + c_2sw_4^2\big)$. The remaining equations, together with $w_4^3 = 1 + c_2^2$ and the definitions of c_3, c_4, are six polynomials in $(c_1, \dots, c_4, w_4, s, w_1)$ of degrees $11, 15, 20, 6, 11, 3$; they are written out by the function `severi_system` of the script. The condition $c_1w_4 \neq 0$ is imposed with a Rabinowitsch variable, and s, w_1, w_4 are eliminated.

C Boundary geometry and dimension counts

Let $\tilde{S}$ be $\hat{E}$ blown up at the triple point p of the type-IV fibre, with exceptional curve E_p. The boundary of S in $\tilde{S}$ is $D = O + \Theta_0' + \Theta_1' + \Theta_2' + E_p$, with $O^2 = -1$, $\Theta_i'^2 = -3$, $E_p^2 = -1$, $O \cdot \Theta_0' = 1$ and $E_p \cdot \Theta_i' = 1$. The pole orders of the coordinate functions along the boundary components are:

	O	Θ_0'	Θ_1'	Θ_2'	E_p
x	0	1	1	1	3
Y	2	2	1	1	4
z	3	3	2	2	6

These follow from the Weierstrass scaling at $t = 1/x$, the A_2 resolution at the node of the type-IV fibre, and the blow-up at p. The polar divisor of z is $3O + 3\Theta_0' + 2\Theta_1' + 2\Theta_2' + 6E_p$; solving $M\,a = b$ for the intersection matrix M of D and $b = (0,0,0,0,1)$ gives exactly this vector, and $L(\cdot) = \langle 1, x, x^2, Y, z\rangle$ has dimension $5 = 1 + (C^2 - K\cdot C)/2$ for $C^2 = 6$, $K\cdot C = -2$. This is the linear system of hyperplane sections of Section 4; arithmetic genus 3 agrees with the Newton polygon count.

A $\mathbb{Q}$-rational curve with $\deg x = 3$ has class G-invariant in $\mathrm{Pic}(S)$, hence trivial (Proposition 2.3), so it is numerically a combination of boundary components: either the hyperplane class above, with a transversal point at infinity on E_p (expected dimension $-K\cdot C - 1 = 1$ for rational members, as found), or a class $(b_O; 3, 0, 0)$ meeting Θ_0' with multiplicity 3 and O with multiplicity $b_O = \nu - 3$ at a single point, where the requirement that all intersection points coincide costs $b_O + 2$ conditions, giving expected dimension $-b_O$. This heuristic count motivates the expectation that the cases $\nu \geq 4$ are empty; it is not a proof.

References

[1] B. Grechuk, *Polynomial Diophantine Equations: A Systematic Approach*, Springer, 2024, doi:10.1007/978-3-031-62949-5.

[2] B. Grechuk, *A systematic approach to Diophantine equations: open problems*, arXiv:2404.08518.

[3] K. Oguiso and T. Shioda, The Mordell–Weil lattice of a rational elliptic surface, *Comment. Math. Univ. St. Pauli* **40** (1991), 83–99.

[4] T. Shioda, On the Mordell–Weil lattices, *Comment. Math. Univ. St. Pauli* **39** (1990), 211–240.

[5] C. L. Siegel, Über einige Anwendungen diophantischer Approximationen, *Abh. Preuss. Akad. Wiss.* (1929), no. 1.

[6] H. Davenport, On $f^3(t) - g^2(t)$, *Norske Vid. Selsk. Forh.* **38** (1965), 86–87.

[7] W. W. Stothers, Polynomial identities and Hauptmoduln, *Quart. J. Math. Oxford* (2) **32** (1981), 349–370.

[8] R. C. Mason, *Diophantine Equations over Function Fields*, LMS Lecture Note Ser. 96, Cambridge Univ. Press, 1984.

[9] U. Zannier, On Davenport's bound for the degree of $f^3 - g^2$ and Riemann's existence theorem, *Acta Arith.* **71** (1995), 107–137.

[10] W. Decker, G.-M. Greuel, G. Pfister, H. Schönemann, *Singular 4-3-2 — A computer algebra system for polynomial computations*, 2022.

[11] The PARI Group, *PARI/GP*, Univ. Bordeaux.